MATHS MANTHAN
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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 12** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Case Study Question 12(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer questions of 2 marks each.
4. **Section C** comprises 2 Short Answer questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS questions of 5 marks each.
6. **Section E** comprises 1 Applied Case Study question of 5 marks with internal choice.
7. Use of calculators is not permitted. Take $\sqrt{3} = 1.732$ unless stated otherwise.

SECTION A: Multiple Choice & Assertion-Reason (5 × 1 = 5 Marks)

Q1. The angle of elevation of the top of a cliff 100 m high from a point directly at its base is: [1]

(A) $0°$ (B) $45°$ (C) $90°$ (D) undefined

Q2. If the angle of elevation of a cloud from a point h metres above a lake is θ, then the angle of depression of its reflection in the lake is: [1]

(A) equal to θ (B) greater than θ (C) less than θ (D) $90° - \theta$

Q3. The shadow of a tower is 30 m long when the sun's elevation is $30°$. The length of the shadow when the sun's elevation is $60°$ is: [1]

(A) 10 m (B) 15 m (C) $10\sqrt{3}$ m (D) 20 m

Q4. A man on the deck of a ship, 10 m above water level, observes that the angle of elevation of the top of a cliff is $60°$ and the angle of depression of the base is $30°$. The distance of the cliff from the ship is: [1]

(A) $10\sqrt{3}$ m (B) $20\sqrt{3}$ m (C) 30 m (D) 10 m

Q5. **Assertion (A):** In a right-angled triangle formed by a vertical pole, its shadow, and the sun's rays, the hypotenuse represents the line of sight from the tip of the shadow to the sun.
Reason (R): The line of sight is defined as the line drawn from the observer's eye to the point being observed. [1]

(A) Both (A) and (R) are true; (R) explains (A).
(B) Both are true; (R) does not explain (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

SECTION B: Very Short Answer Questions (2 × 2 = 4 Marks)

Q6. From a point P on the ground, the angle of elevation of the top of a 10 m tall building is $30°$. A flag is hoisted at the top of the building and the angle of elevation of the top of the flagstaff from P is $45°$. Find the length of the flagstaff. [Take $\sqrt{3} = 1.732$] [2]

Q7. A steel wire is attached to the top of an electric pole of height 14 m to anchor it firmly to the ground. If the wire is inclined at $45°$ to the horizontal, find the length of the wire. [Take $\sqrt{2} = 1.414$] [2]

SECTION C: Short Answer Questions (2 × 3 = 6 Marks)

Q8. An observer on the top of a cliff 200 m high observes two cargo ships moving away from the cliff in the sea in the same direction. If the angles of depression of the ships change from $45°$ to $30°$, find the distance between the two ships. [Take $\sqrt{3} = 1.732$] [3]

Q9. A car is approaching a tower at a uniform speed. The angle of elevation of the top of the tower from the car changes from $30°$ to $60°$ in 12 minutes. How much more time will the car take to reach the base of the tower? [3]

SECTION D: Long Answer / HOTS Questions (2 × 5 = 10 Marks)

Q10. The angle of elevation of a stationary cloud from a point h metres above the surface of a lake is θ and the angle of depression of its reflection in the lake is ϕ. Prove that the height of the cloud above the lake surface is: [5]

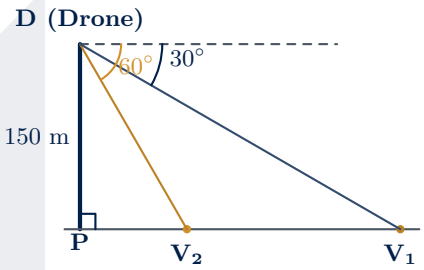
$$H = h\left(\frac{\tan\phi + \tan\theta}{\tan\phi - \tan\theta}\right)$$

If $h = 250$ m, $\theta = 30°$, and $\phi = 60°$, calculate the numerical height of the cloud above the lake.

Q11. An aeroplane flying at a constant altitude of 3600 m is observed simultaneously from two ground radar stations A and B situated 4800 m apart on level ground. If the aeroplane is between the two stations in the same vertical plane and the angle of elevation from station A is 60°, calculate the distance of the aeroplane from station B. [Take $\sqrt{3} = 1.732$] [5]

SECTION E: Applied Case Study (1 × 5 = 5 Marks)

Q12. CASE STUDY: Tactical Drone Aerial Reconnaissance and Border Security



A high-tech border surveillance quadcopter drone hovers stationary at a constant operational altitude of 150 m directly above a security checkpoint P. The drone's long-range electro-optical camera system monitors approaching vehicle traffic along a straight perimeter access road.
The surveillance operator spots an approaching vehicle at position V_1 at an angle of depression of 30°. Exactly 3 minutes later, the vehicle reaches position V_2 where its angle of depression increases to 60°.

Based on the above mathematical scenario, answer the following questions:

(i) Find the initial horizontal distance of the vehicle from the checkpoint (PV_1). [Take $\sqrt{3} = 1.732$] [1]

(ii) Find the horizontal distance of the vehicle from the checkpoint at the second observation (PV_2). [2]

(iii) Calculate the average speed of the approaching vehicle in km/h during this 3-minute observation interval. [2]

— OR —

If the vehicle continues to travel at this uniform speed, how many more seconds will it take to reach the security checkpoint P? [2]

★★★ END OF QUESTION PAPER · ALL THE BEST ★★★